CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 8: Logarithms

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write in logarithmic form: $5^3 = 125$ and $10^{-3} = 0.001$.

2. Find $\log_3 243$ and $\log_4 \dfrac{1}{64}$.

3. Find x if $\log_x 125 = 3$ and if $\log_{16} x = \dfrac{3}{4}$.

4. Express $\log \dfrac{x^3}{y^2 z}$ in terms of $\log x$, $\log y$, $\log z$.

5. Write as a single logarithm: $\log 6 + 2\log 5 - \log 15$.

6. Evaluate $\dfrac{\log 32}{\log 4}$.

7. Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 18$ and $\log 25$.

8. Solve $\log(x+1) + \log(x-1) = \log 24$.

9. Solve $\log(3x+1) = 2$.

10. If $\log(x-y) = \log x - \log y$, express x in terms of y.

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Answer Key

1. $\log_5 125 = 3$ and $\log_{10} 0.001 = -3$
2. 5 and -3
3. $x = 5; x = 8$
4. $3\log x - 2\log y - \log z$
5. $\log 10 = 1$
6. $\frac{5}{2}$
7. 1.2552 and 1.3980
8. $x = 5$
9. $x = 33$
10. $x = \frac{y^2}{y-1}$

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